

STANA DE VALE, Romania

WMO: 151180

Lat: 46.6899N

Lon: 22.6233E

Elev: 3639

StdP: 12.86

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.6	(c) 3.1	(d) -4.1	(e) 5.0	(f) 0.6	(g) 0.5	(h) 6.4	(i) 3.6	(j) 13.1	(k) 29.3	(l) 8.4	(m) 28.1	(n) 0.8	(o) 140	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 23.1	(c) 78.5	(d) 62.9	(e) 75.1	(f) 62.7	(g) 72.0	(h) 61.3	(i) 65.6	(j) 74.3	(k) 64.0	(l) 72.2	(m) 62.6	(n) 70.7	(o) 2.5	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.9	98.3	68.5	61.1	92.2	67.4	59.5	86.9	66.0	32.5	75.4	31.3	72.1	30.1	70.6	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.0	(b) 6.0	(c) 4.8	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 9.0	(b) 6.0	(c) 4.8	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	22.9	27.6	31.9	40.0	49.2	55.5	57.3	58.5	50.6	42.8	35.2	26.9		
	DBStd	14.18	8.87	8.16	7.17	8.35	5.97	5.88	5.17	5.34	7.36	7.57	8.44	8.19		
	HDD50	3939	840	627	562	308	86	16	7	4	86	240	447	716		
	HDD65	8555	1305	1047	1027	749	491	287	243	208	434	688	895	1181		
	CDD50	876	0	0	0	9	60	182	233	269	104	17	2	0		
	CDD65	16	0	0	0	0	0	3	4	8	1	0	0	0		
	CDH74	428	0	0	0	0	7	46	138	215	22	0	0	0		
	CDH80	56	0	0	0	0	1	0	12	42	1	0	0	0		
	WSAvg	1.3	1.0	1.3	1.4	1.5	1.4	1.3	1.3	1.2	1.2	1.2	1.4	1.1		
	PrecAvg	41.7	2.6	2.3	2.7	3.2	4.0	5.0	4.9	3.3	3.9	3.3	3.1	3.4		
	PrecMax	56.6	5.5	6.2	7.9	5.1	6.9	8.3	9.2	8.9	10.7	8.8	7.0	7.6		
(15) Precipitation	PrecMin	30.6	1.1	0.5	0.8	0.7	1.5	1.3	2.0	0.2	0.6	0.2	0.0	0.2		
	PrecStd	6.8	1.1	1.3	1.9	1.2	1.5	1.7	2.2	1.9	2.2	2.0	1.4	2.0		
	DB	42.5	49.9	57.4	68.9	73.7	78.1	81.7	84.2	77.0	68.5	59.2	46.0			
	MCWB	38.8	41.0	44.6	53.2	58.2	63.5	63.0	65.7	61.6	55.2	48.4	41.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	39.1	45.4	51.9	64.4	69.1	75.1	78.1	79.5	71.8	64.7	55.3	42.0			
	2% MCWB	37.2	39.5	42.1	50.8	57.1	63.7	62.6	63.1	59.3	54.0	47.7	39.0			
	5% DB	36.7	42.2	47.3	59.9	65.6	72.2	73.8	74.9	68.0	60.2	51.4	39.2			
	5% MCWB	35.2	37.8	40.0	48.8	55.1	62.2	62.0	62.7	57.9	52.3	46.0	36.9			
	10% DB	34.1	39.1	43.5	54.9	62.1	68.9	70.2	71.0	64.0	55.6	47.3	36.4			
	10% MCWB	33.1	36.3	37.9	45.6	53.7	60.4	60.4	61.6	56.9	50.2	44.1	35.2			
	0.4% WB	40.3	43.8	46.9	54.6	60.2	67.8	66.3	67.8	62.8	56.7	50.3	42.7			
	0.4% MCDB	41.6	47.4	55.8	66.6	68.8	74.6	74.6	78.9	72.6	63.3	56.1	45.1			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	37.6	40.2	43.3	52.1	58.2	65.2	64.7	65.7	60.9	55.3	48.4	40.2			
	2% MCDB	38.6	44.0	50.5	63.4	66.3	72.4	73.0	75.7	69.5	62.4	54.2	42.1			
	5% WB	35.5	38.4	40.8	49.5	56.5	63.2	63.3	64.0	59.0	53.3	46.6	37.2			
	5% MCDB	36.4	41.4	46.7	58.8	64.2	70.9	71.5	72.5	66.6	60.4	51.1	39.1			
	10% WB	33.2	36.4	38.5	46.6	54.5	61.0	61.7	62.6	57.2	50.7	44.7	35.2			
	10% MCDB	34.0	38.7	42.7	53.3	61.1	68.0	69.4	70.6	63.5	55.2	47.9	36.4			
	MDBR	16.7	17.3	18.5	21.1	20.2	21.2	22.0	23.1	19.4	19.8	14.9	13.7			
	5% DB	17.2	20.5	24.9	29.7	26.8	26.1	29.2	29.7	26.4	28.0	20.7	14.6			
(36) Mean Daily Temperature Range	MCWBR	14.2	14.4	16.2	16.6	16.1	15.2	16.0	15.2	15.0	17.7	13.2	11.6			
	MCDBR	14.3	17.2	23.2	26.7	24.8	24.3	25.6	26.3	25.2	27.1	18.7	13.5			
	MCWBR	12.0	12.6	15.8	15.7	15.7	15.2	15.1	15.0	14.9	17.6	12.6	10.9			
	MCDBR	12.0	12.6	15.8	15.7	15.7	15.2	15.1	15.0	14.9	17.6	12.6	10.9			
(40) Clear-Sky Solar Irradiance	taub	0.292	0.319	0.358	0.405	0.391	0.387	0.402	0.405	0.373	0.343	0.317	0.294			
	taud	2.362	2.313	2.260	2.155	2.251	2.290	2.249	2.237	2.307	2.387	2.407	2.399			
	Ebn at Noon	253	268	272	269	277	279	272	266	265	256	241	239			
	Edh at Noon	23	30	36	44	42	41	42	41	35	27	22	20			
(44) All-Sky Solar Radiation	RadAvg	373	627	998	1384	1674	1845	1807	1667	1180	801	456	291			
	RadStd	53	84	146	142	162	149	160	158	135	97	63	40			

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)	+1.28	N/A	N/A	N/A	+1.80	N/A	N/A	-234	-355	+207	+104

Nomenclature: See separate page